

10.14 (a) Following Section 10.9, we have

$$\vec{E}_i = E_0 (\vec{e}_1 \sin \beta + \vec{e}_2 \cos \beta) e^{ikz}, \quad (\vec{n} \times \vec{E}_i)_{z=0} = E_0 (-\vec{e}_1 \cos \beta + \vec{e}_2 \sin \beta)$$

and the scattered field is

$$\vec{E}(\vec{r}) = \frac{ie^{ikr} E_0}{2\pi r} \vec{k} \times \int (-\vec{e}_1 \cos \beta + \vec{e}_2 \sin \beta) e^{-i\vec{k} \cdot \vec{r}'} da'$$

For arbitrary $\vec{k} = k(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, we have

$$\vec{k} \times (-\vec{e}_1 \cos \beta + \vec{e}_2 \sin \beta) = (-\vec{e}_1 \cos \theta \sin \beta - \vec{e}_2 \cos \theta \cos \beta + \vec{e}_3 \sin \theta \sin(\phi + \beta)) k$$

$$\vec{k} \cdot \vec{r}' = k(\sin \theta \cos \phi x' + \sin \theta \sin \phi y'),$$

$$\text{and } \vec{E}(\vec{r}) = \frac{ie^{ikr} E_0 k}{2\pi r} (-\vec{e}_1 \cos \theta \sin \beta - \vec{e}_2 \cos \theta \cos \beta + \vec{e}_3 \sin \theta \sin(\phi + \beta))$$

$$\begin{aligned} & \int_{-a/2}^{a/2} \exp[-ikx' \sin \theta \cos \phi] dx' \int_{-b/2}^{b/2} \exp[-iky' \sin \theta \sin \phi] dy' \\ &= \frac{ie^{ikr} E_0 k}{2\pi r} \left(-\vec{e}_1 \cos \theta \sin \beta - \vec{e}_2 \cos \theta \cos \beta + \vec{e}_3 \sin \theta \sin(\phi + \beta) \right) \\ & \quad \frac{\sin(k a \sin \theta \cos \phi / 2)}{k \sin \theta \cos \phi / 2} \frac{\sin(k b \sin \theta \sin \phi / 2)}{k \sin \theta \sin \phi / 2} \end{aligned}$$

The power per unit solid angle is

$$\frac{dP}{d\Omega} = \frac{E_0^2}{2Z_0} \frac{k^2}{4\pi^2} (\cos^2 \theta + \sin^2 \theta \sin^2(\phi + \beta)) \left| \frac{\sin(\frac{1}{2} k a \sin \theta \cos \phi)}{\frac{1}{2} k \sin \theta \cos \phi} \frac{\sin(\frac{1}{2} k b \sin \theta \sin \phi)}{\frac{1}{2} k \sin \theta \sin \phi} \right|^2$$

(b) For scalar scattering calculation, take $\psi(\vec{r}) = I_0 e^{ikz}$, then by Eq. (10.108),

$$\psi(\vec{r}) = -\frac{e^{ikr}}{4\pi r} \int e^{-i\vec{k} \cdot \vec{r}'} [ik + ik \cos \theta] I_0 da'$$

$$= -\frac{ie^{ikr} I_0 k}{4\pi r} (1 + \cos \theta) \int e^{-i\vec{k} \cdot \vec{r}'} da'$$

$$= -\frac{ie^{ikr} I_0 k}{4\pi r} (1 + \cos \theta) \frac{\sin(\frac{1}{2} k a \sin \theta \cos \phi)}{\frac{1}{2} k \sin \theta \cos \phi} \frac{\sin(\frac{1}{2} k b \sin \theta \sin \phi)}{\frac{1}{2} k \sin \theta \sin \phi}$$

$$\text{and } \frac{dp}{dn} = I_0 \frac{k^2}{4\pi^2} \left(\frac{1 + \cos\theta}{2} \right)^2 \left| \frac{\sin\left(\frac{1}{2}ka\sin\theta\cos\phi\right)}{\frac{1}{2}k\sin\theta\cos\phi} \frac{\sin\left(\frac{1}{2}kb\sin\theta\sin\phi\right)}{\frac{1}{2}k\sin\theta\sin\phi} \right|^2$$